

| Question | Marking Guidance | Mark              | Comments                                                                       |
|----------|------------------|-------------------|--------------------------------------------------------------------------------|
| 03.1     | 2/two;           | 1<br>(1 x<br>AO2) | If no number is present, accept <i>Gastrophysa</i> and <i>Chrysolina</i> alone |

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| 03.2 | <p><b>Max 2 marks for MP1-5 and Max 1 mark for MP6-8</b></p> <p><b>Reasons for capture</b></p> <ol style="list-style-type: none"> <li>(So beetles) have emerged (from the soil)</li> </ol> <p><b>OR</b></p> <p>(So beetles) are not still in burrows</p> <p><b>OR</b></p> <p>(So beetles) are not under the soil/ground;</p> <ol style="list-style-type: none"> <li>(To allow beetles to) evenly/randomly distribute (before they burrow underground);</li> <li>To allow <b>time</b> for (re)capture (before they burrow underground);</li> <li>To not disturb mating/egg laying (in April);</li> <li>To ensure no population changes/increase (due to production of eggs/larvae);</li> </ol> <p><b>Reason for identification</b></p> <ol style="list-style-type: none"> <li>(All three beetles are) same colour/green;</li> <li>(All three beetles are a) similar length;</li> <li>Only <i>C. graminis</i>/tansy beetle (population size being) investigated</li> </ol> <p><b>OR</b></p> <p>Only investigated one/correct species</p> <p><b>OR</b></p> <p>Did not investigate (the) other/wrong species;</p> | 3 max<br>(3 x<br>AO3) | 7. Accept size |
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| Question | Marking Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Mark                       | Comments |
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| 03.3     | <p>Correct answer of 1215 = <b>2 marks</b>;;</p> <p>Evidence of 810 (not calculated total number in 2<sup>nd</sup> sample) = <b>1 mark</b></p> <p><b>OR</b></p> <p>Evidence of 608 (divided by unmarked rather than marked) = <b>1 mark</b></p> <p><b>OR</b></p> <p>Evidence of <math>\frac{150\ 660}{124}</math> (correct working but incorrect or no answer) = <b>1 mark</b></p> <p><b>OR</b></p> <p>Evidence of <math>\frac{405 \times 372}{124}</math> (correct working but incorrect or no answer) = <b>1 mark</b></p> <p><b>OR</b></p> <p>Evidence of <math>\frac{405}{n} \times \frac{124}{372}</math> (correct working but incorrect or no answer) = <b>1 mark</b>;</p> | <p>2<br/>(2 x<br/>AO2)</p> |          |

| Question | Marking Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Mark               | Comments                                                                                                          |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------|
| 03.4     | <ol style="list-style-type: none"> <li>Collected samples during the second week of August</li> </ol> <p><b>OR</b></p> <p>Collected samples at the same time of year;</p> <ol style="list-style-type: none"> <li>Collected samples from the same location/area/habitat;</li> <li>Used the same method;</li> <li>Used a different mark (each year);</li> <li>Marking does not affect survival/predation/behaviour/recapture</li> </ol> <p><b>OR</b></p> <p>Mark is not toxic</p> <p><b>OR</b></p> <p>Mark is not removed (before second capture);</p> | 2 max<br>(2 x AO3) | <ol style="list-style-type: none"> <li>Accept any correct named part of the method used being the same</li> </ol> |

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